MIL-T-37619A
22 September 1977
SUPERSEDING
MIL-T-37619(DSA-DM)
26 July 1976

MILITARY SPECIFICATION

TRIMETHOPRIM AND SULFAMETHOXAZOLE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Trimethoprim and Sulfamethoxazole Tablets.

1.2 Classification. The tablets covered by this specification shall be of the following sizes and strengths (see 6.2):

Size 1 - 100's
Size 2 - 500's
Size 3 - Individually Sealed, 100's
Strength 1 - 400 mg of Sulfamethoxazole and 80 mg of Trimethoprim
Strength 2 - 800 mg of Sulfamethoxazole and 160 mg of Trimethoprim

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Trimethoprim and Sulfamethoxazole Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

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S2. Classification. Shall be type I.

S5.3 The following is added to this paragraph:

Strength 1 shall be tablets containing 400 mg of sulfamethoxazole and 80 mg of trimethoprim per tablet, within the designated assay limits for the tablets.

Strength 2 shall be tablets containing 800 mg of sulfamethoxazole and 160 mg of trimethoprim per tablet, within the designated assay limits for the tablets.

The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of sulfamethoxazole and trimethoprim when assayed as specified in 4.3.1.1 and 4.3.1.2.

The tablets shall comply with the identification tests specified in 4.3.2.

3.2 Ingredients.

3.2.1 Sulfamethoxazole. The sulfamethoxazole used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.2.2 Trimethoprim. The trimethoprim used in the manufacture of the tablets shall comply with the following requirements:

3.2.2.1 Identification. Shall comply with the identification tests specified in 4.3.3.1.

3.2.2.2 Melting range. Shall melt between 199° and 203°C using the USP method for Class I material.

3.2.2.3 Loss on drying. Shall lose not more than 0.5 percent of its weight when dried for 4 hours at 105°C using the USP method.

3.2.2.4 Residue on ignition. Shall be not more than 0.2 percent when determined using the USP method.

3.2.2.5 Heavy metals. Shall contain not more than 20 ppm heavy metals when determined by the USP procedure (Method II).

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3.2.2.6 Assay. Shall contain not less than 98.5 percent and not more than 101.0 percent trimethoprim when assayed as specified in 4.3.3.2.

3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Tests.

4.3.1 Assay of finished tablets.

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4.3.1.1 Sulfamethoxazole.

Grind 20 tablets to a fine powder. Accurately weigh a portion of the powder, equivalent to about 750 mg of sulfamethoxazole, into a 250 ml beaker. Add 20 ml of glacial acetic acid, 40 ml of purified water and 15 ml of concentrated hydrochloric acid. Titrate potentiometrically with 0.1 M sodium nitrite, using a platinum/calomel electrode system. Each ml of 0.1 M sodium nitrite is equivalent to 25.33 mg of sulfamethoxazole.

Calculation.

$$\frac{(\text{ml } NaNO_2) \times (\text{M } NaNO_2) \times (253.3) \times (\text{Avg. Tab. Wt. in mg})}{\text{Sample Wt. in mg}} = \text{mg of sulfamethoxazole per tablet}$$

4.3.1.2 Trimethoprim

Reagents

Phosphate Buffer, pH 5.3: Transfer 38 grams of monobasic sodium phosphate monohydrate and 3.8 grams of dibasic sodium phosphate heptahydrate to a 1 liter graduate. Dissolve in and dilute to volume with purified water. Adjust the pH, if necessary.

Bromcresol Green Solution: Dissolve 250 mg of tetrabromo-m-cresolsulfonphthalein sodium in 250 ml of phosphate buffer.

N, N-Dimethylformamide (DMF): reagent grade

Chloroform: spectrophotometric grade

Methanol: reagent grade

Blank

Dilute 25.0 ml of DMF to 100 ml with methanol. Subdilute 10.0 ml of this solution to 100 ml with chloroform (Blank Solution).

Standard

Accurately weigh about 60 mg of a trimethoprim reference standard into a 100 ml volumetric flask. Dissolve in 25.0 ml of DMF and dilute to volume with methanol. Subdilute 10.0 ml of this solution to 100 ml with chloroform (Standard Solution).

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Sample

Accurately weigh an amount of finely ground sample equivalent to about 60 mg of trimethoprim into a 100 ml volumetric flask. Add 25.0 ml of DMF and shake mechanically for 5 minutes. Dilute to volume with methanol. Subdilute 10.0 ml of this solution, to 100 ml with chloroform (Sample Solution).

Procedure

Pipet the following volumes (ml) of the indicated solutions into separate 125 ml separators:

	Blank	Standard	Sample
Blank Solution	10	-	-
Standard Solution	-	10	-
Sample Solution	-	-	10
Phosphate Buffer	25	25	25
Bromcresol Green Solution	5	5	5

Extract the contents of each separator with 3 x 25 ml portions of chloroform, shaking each separator for 1 minute and collecting the extracts in separate 100 ml volumetric flasks. Dilute to volume with chloroform and using a suitable spectrophotometer measure the absorbance of each solution against the Blank in the reference cell at 420 nm.

Calculation.

$$\frac{A_{\mu} \times \text{Std. Wt. (mg)} \times \text{Avg. Tab. Wt. (mg)}}{A_S \times \text{Sample Wt. (mg)}} = \text{mg of trimethoprim per tablet}$$

Where
A_{μ} = absorbance of sample solution at 420 nm
A_S = absorbance of standard solution at 420 nm

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4.3.2 Identification of finished tablets.

4.3.2.1 Identity Test A

To a portion of finely ground tablets, equivalent to about 100 mg of sulfamethoxazole, add 2-3 ml of approx. 0.1N hydrochloric acid and 3 ml of 1 percent aqueous sodium nitrite. Then add 1 ml of freshly prepared bethanaphthol TS: an orange precipitate forms (perform a blank determination).

4.3.2.2 Identity Test B

Triturate a portion of finely ground tablets, equivalent to about 40 mg of trimethoprim, with 5-10 ml of dioxane. Filter the triturate and add 2 ml of a saturated solution of chloranil in dioxane to the filtrate: a brown-red color appears on standing (perform a blank determination).

4.3.3 Tests for Ingredients.

* 4.3.3.1 Identification of trimethoprim

Prepare a 1 percent solution of trimethoprim in chloroform. Place the solution into a 0.4 mm sodium chloride cell. Record the infrared spectrum of the solution using a suitable infrared spectrophotometer. The spectrum obtained shall agree qualitatively with that of a trimethoprim reference material similarly prepared and measured on the same spectrophotometer. As an alternate, a 0.3 percent solution of the sample in chloroform may be prepared and this solution may be placed into a 1.0 mm sodium chloride cell.

4.3.3.2 Assay for trimethoprim.

Accurately weigh about 0.9 gram of trimethoprim into a 250 ml beaker and dissolve in about 100 ml of acetic anhydride glacial acetic acid (90:10). Titrate with 0.1N perchloric acid in glacial acetic acid either potentiometrically, using a glass-calomel electrode system, or visually to give a green endpoint, using crystal violet TS indicator. Perform a blank titration and make any necessary volume correction. One ml of 0.1N perchloric acid is equivalent to 29.03 mg of trimethoprim.

Calculation.

$$\frac{(\text{ml } HClO_4) \times (\text{N } HClO_4) \times (0.2903) \times 100}{\text{Sample Wt. (gram)}} = \text{percent trimethoprim}$$

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4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 100 tablets of size 1 or one bottle containing 500 tablets of size 2 or one box containing 100 tablets of size 3, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container: size 1 and size 2. The immediate container for size 1 and size 2 shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Immediate container: size 3. The immediate container for size 3 shall be a commercially available blister or strip packaging considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.3 Unit packaging. At the option of the contractor, each unit of size 1 or size 2 may be packaged as specified in 5.2.5 of PPP-C-186. Each unit of size 3 shall be packaged as specified in 5.2.5 of PPP-C-186.

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5.2.2.4 Intermediate packaging. Twelve (12) units of issue of size 1 or ten (10) units of issue of size 3 shall be packaged as specified in 5.2.7.1 of PPP-C-186. Twelve (12) units of issue of size 2 shall be packaged as specified in 5.2.8 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Forty-eight (48) units of size 1 or twenty-four (24) units of issue of size 2 or one hundred (100) units of issue of size 3 shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Forty-eight (48) units of size 1 or one hundred (100) units of issue of size 3 shall be packed as specified in 5.4.3.1.1 of PPP-C-186. Twenty-four (24) units of issue of size 2 shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

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5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied for size 1 or size 2, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. In addition, Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)"

* 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-00-148-7164	TRIMETHOPRIN AND SULFAMETHOXAZOLE TABLETS, 500s	Size 2, Strength 1
6505-01-016-1470	TRIMETHOPRIM AND SULFAMETHOXAZOLE TABLETS, 100s	Size 1, Strength 2
6505-01-017-8470	TRIMETHOPRIM AND SULFAMETHOXAZOLE TABLETS, Individually Sealed, 100s	Size 3, Strength 2

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assume no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6505-1973

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-37619A TRIMETHOPRIM AND SULFAMETHOXAZOLE TABLETS

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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